REVIEW OF IMPORTANT RESULTS

- **Proposition 32**: A graph with at least two vertices is bipartite if and only if it has no odd cycle.
- **Remark**: The above is an instance of TONCAS (*T*he *O*bvious *N*ecessary *C*ondition *A*lso *S*ufficient). TONCAS is a very common occurrence in Graph Theory.
- **Proposition 33:** If every vertex of graph G has degree at least 2, then G contains a cycle.

Eulerian Circuits

Some Definitions:

- A graph is ***Eulerian*** if it has a ***closed*** trail passing through all the edges exactly once. For convenience, a graph consisting of trivial components is regarded as Eulerian.
- We call a closed trail a ***circuit*** when we do not specify the first vertex but keep the list in cyclic order.
- An ***Eulerian circuit*** or ***Eulerian trail*** in a graph is a circuit or trail containing all the edges.
- An ***even graph*** is a graph with vertex degrees all even.
- A vertex is ***odd*** [*even*] when its degree is odd [even].

Theorem 6 (Euler – 1735): A graph G is Eulerian if and only if it has at most one nontrivial component and its vertices all have even degree.

even graph

- The Seven Bridges of Königsberg is a historically notable problem in mathematics. Its negative resolution by Leonhard Euler in 1735 laid the foundations of graph theory and prefigured the idea of topology.

Not Eulerian
not even

- The city of Königsberg in Prussia (now Kaliningrad, Russia) was set on both sides of the Pregel River, and included two large islands which were connected to each other and the mainland by seven bridges.

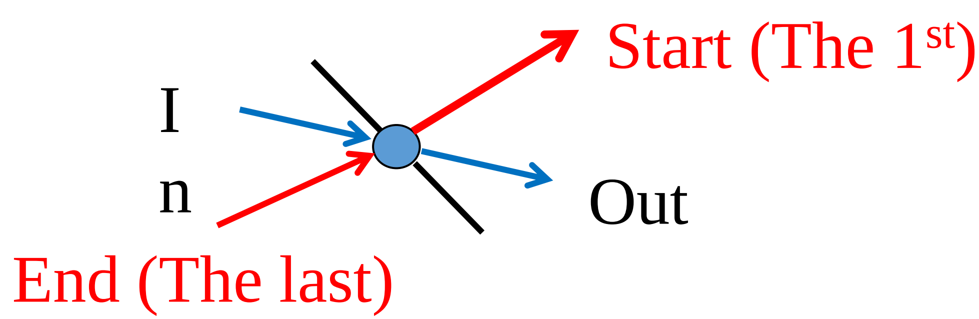
- The problem was to find a walk through the city that would cross each bridge once and only once.

Theorem 6 (Euler – 1735): A graph *G* is Eulerian if and only if it has at most one nontrivial component and its vertices all have even degree.

Proof: (*Necessity*)

TONCAS

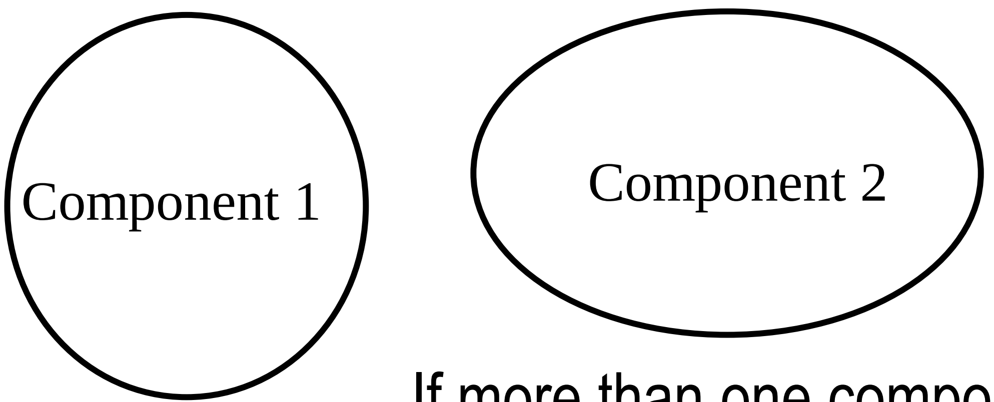
- Suppose that ***G*** has an Eulerian circuit ***C***
- Each passage of ***C*** through a vertex uses two incident edges
- And the first edge is paired with the last at the first vertex
- Hence every vertex has even degree



Theorem 6: A graph G is Eulerian if and only if it has at most one nontrivial component and its vertices all have even degree.

Proof: (*Necessity*)

- Also, two edges can be in the same trail only when they lie in the same component, so there is at most one nontrivial component.



If more than one components, can't walk across the graph

Theorem 6 (continued): A graph *G* is Eulerian if and only if it has at most one nontrivial component and its vertices all have even degree

Proof: (*Sufficiency* 1/3)

- Assuming that the condition holds, we obtain an Eulerian circuit using induction on the number of edges, ***m*** ← Strong Induction
- Basis step: ***m***= 0. A closed trail consisting of one vertex suffices →

Theorem 6 (continued): A graph G is Eulerian if and only if it has at most one nontrivial component and its vertices all have even degree.

Proof: (*Sufficiency* 2/3)

- Induction step: $m > 0$.
  - When even degrees, each vertex in the nontrivial component of G has degree at least 2.
  - The nontrivial component has a cycle C (Proposition 33)
  - Let H be the graph obtained from G by deleting $E(C)$.
  - Since C has 0 or 2 edges at each vertex, each component of H is also an even graph.
  - Since each component is also connected and has fewer than m edges, we can apply the induction hypothesis to conclude that each component of H has an Eulerian circuit. →

Theorem 6: A graph G is Eulerian if and only if it has at most one nontrivial component and its vertices all have even degree.

Proof: (*Sufficiency* *3/3*)

- Induction step: $m > 0$. (continued)
  - To combine these into an Eulerian circuit of G, we traverse C, but when a component of H is entered ***for the first time*** we detour along an Eulerian circuit of that component.
  - This circuit ends at the vertex where we began the detour. When we complete the traversal of C, we have completed an Eulerian circuit of G.

Corollaries to Euler's Theorem

Corollary 6.1: Let G be a connected graph with exactly two vertices of odd degree, say u and v. Then G has an Eulerian trail that starts at u and ends at v.

Proof: The proof is a variant of the proof for the Theorem. ***Left as an exercise.***

Definition: A **decomposition** of a graph is a list of subgraphs such that every edge of G belongs to exactly one of the subgraphs in the list. Results and problems of interest typically take the subgraphs to be of some common family, e.g. paths of a given length, cycles, etc. The following is a useful example of this kind of result.

Corollary 6.2: Every connected nontrivial even graph decomposes into cycles.

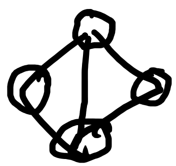
Proof:

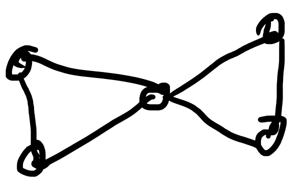
- In Proposition 33 : It is noted that every even nontrivial graph has a cycle
  - The deletion of a cycle leaves an even graph
- Thus this proposition follows by induction on the number of edges

Eulerian & Hamiltonian Graphs

Eulerian Circuit = a circuit going thru' all edges (no repetition)

Hamiltonian Cycle = a cycle going thru' all vertices.

 → Hamiltonian not Eulerian

 → Eulerian not Hamiltonian

Eulerian
- Easy N-S condition
- Finding Eulerian circuit – easy, complexity m (or so?)

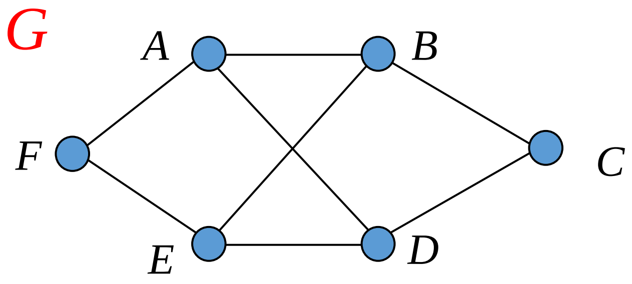
Hamiltonian
- Hard N-S conditions, not easy to verify (will study later)
- Finding Hamiltonian cycle (NP-complete)

Order and Size

- The ***order*** of a graph ***G***, written ***n (G)***, is the number of vertices in ***G***.
- An *n-**vertex graph*** is a graph of order *n*.
- The ***size*** of a graph ***G***, written ***e (G)***, is the number of edges in *G*.
- Notation: For $n \in \mathbb{N}$, $n > 0$, we will use both M_n and [***n***] to indicate the set $\{1,\ldots, n\}$.

Degree

- The **degree** of vertex v in a graph G, $d(v)$ is the number of edges incident to v, except that each loop at v counts twice
- The **maximum degree** is $\Delta(G)$
- The **minimum degree** is $\delta(G)$

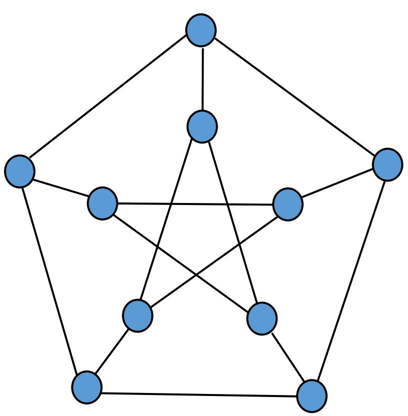


$d(B) = 3,\ d(C) = 2$

$\Delta(G) = 3,\ \delta(G) = 2$

Regular

- ***G*** is *regular* if $\Delta($***G***$) = \delta($***G***$)$
- ***G*** is *k-regular* if the common degree is ***k***.
- The *neighborhood* of ***v***, written $N_g($***v***$)$ or $N($***v***$)$ is the set of vertices adjacent to ***v***.

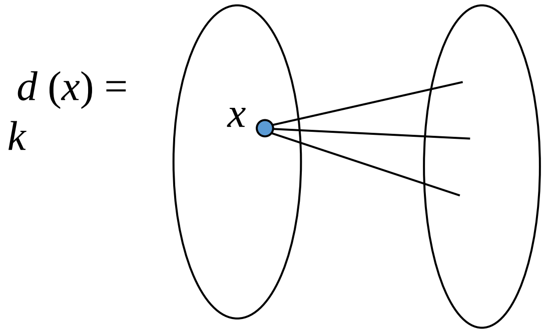


3-*regular*

Proposition 34 : If $k > 0$, then a k-regular bipartite graph has the same number of vertices in each partite set.

Proof:

- Let ***G*** be an ***X,Y*** – bipartite graph.
- Counting the edges according to their endpoints in ***X*** yields $\boldsymbol{e(G) = k\,|X|}$.
- By symmetry, $e(G) = k\,|Y|$, and the result follows.



Extremality Results - 1

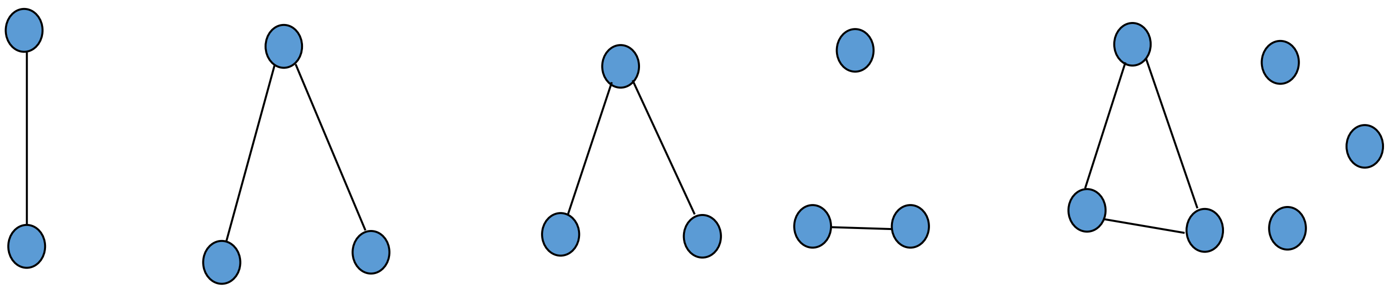
Proposition 35: Every graph with n vertices and k edges has at least $n-k$ components

Proof:

- An n-vertex graph with no edges has n components
- Each edge added reduces this by at most 1
- If k edges are added, then the number of components is at least $n-k$

Proposition 35: Every graph with *n* vertices and *k* edges has at least *n*-*k* components

- Examples:



***n* =2, *k* =1,**
1 component

***n* =3, *k* =2,**
1 component

***n* =6, *k* =3,**
3 components

***n* =6, *k* =3,**
4 components